LAN CONFIGURATION

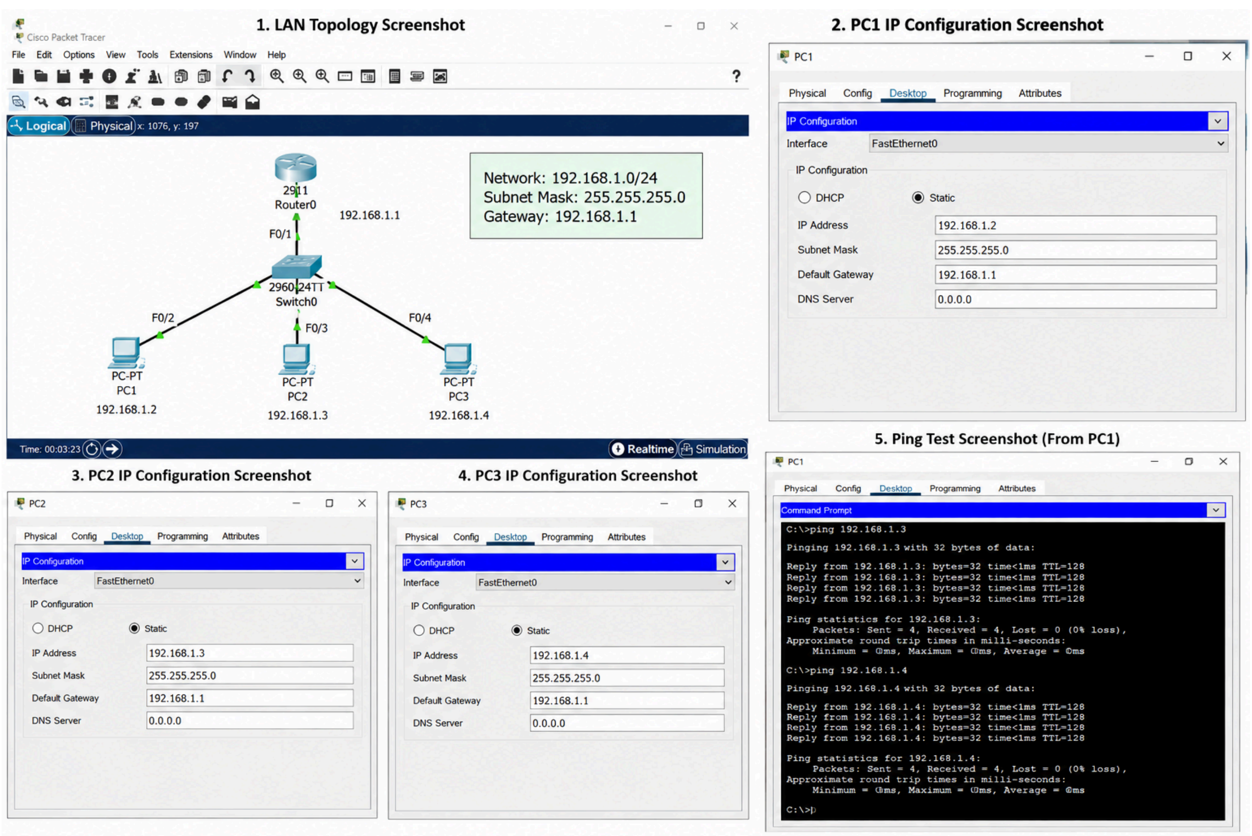
1. Aim

To design and configure a Local Area Network (LAN) using a network switch, router, and multiple computers, assign suitable IP addresses to each device, and verify communication between the devices.

2. Requirements

- Cisco Packet Tracer
- 1 Router
- 1 Cisco 2960 Switch
- 3 PCs
- Ethernet cables
- Computer/Laptop for simulation

3. LAN Topology



4. IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.1.1	255.255.255.0	—
PC1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	192.168.1.3	255.255.255.0	192.168.1.1
PC3	192.168.1.4	255.255.255.0	192.168.1.1

Network Details

- **Network Address:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1
- **Broadcast Address:** 192.168.1.255

5. Configuration Procedure

Step 1: Open Cisco Packet Tracer

Open **Cisco Packet Tracer** and create a new workspace.

Step 2: Add Network Devices

Place the following devices in the workspace:

- One Router
- One Cisco 2960 Switch
- Three PCs

Step 3: Connect the Devices

Connect the three PCs to the switch using **Copper Straight-Through cables**.

Connect the switch to the router using an Ethernet cable.

Step 4: Configure Router

Configure the router interface with:

IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0

Enable the router interface.

Step 5: Configure PC1

Open:

PC1 → Desktop → IP Configuration

Enter:

IP Address : 192.168.1.2
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Step 6: Configure PC2

IP Address : 192.168.1.3
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Step 7: Configure PC3

IP Address : 192.168.1.4
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Step 8: Check Connections

Check the connection indicators in Cisco Packet Tracer. The links should become active and show successful connectivity.

6. Connectivity Testing

Connectivity between the computers can be verified using the **ping command**.

From PC1, open:

PC1 → Desktop → Command Prompt

Enter:

ping 192.168.1.3

Then:

ping 192.168.1.4

If the configuration is correct, successful replies will be received.

Expected Result

Reply from 192.168.1.3
Reply from 192.168.1.4

This confirms that PC1 can communicate with PC2 and PC3 through the LAN.

7. Outcome / Result

The Local Area Network was successfully designed and configured using a router, switch, and three computers. Appropriate IP addresses, subnet masks, and default gateway values were assigned to the devices. Connectivity between the devices was successfully verified using the ping command.

Thus, the LAN was successfully configured and communication between the network devices was established**.**